ENGL220 | SPRING 2026 | DR. NAQAA ABBAS

WALKABILITY IN EDUCATION CITY

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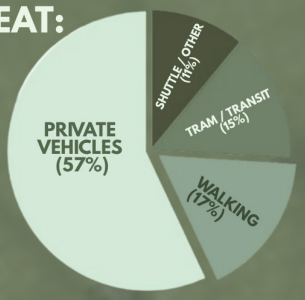
THE PROBLEM

WALKING IN THE HEAT:

- 40˚C+ summer temperatures
- 12 km² mostly unshaded campus
- Only %17 percent actually walk

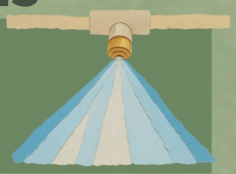


Transportation mode distribution among the Education City population [1]

SOLUTIONS

SOLAR-POWERED MISTING SYSTEMS

Solar-powered nozzles cool the surrounding air.

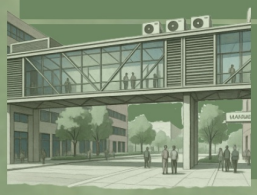


PARK AND PLANT

Native Sidra and Ghaf trees planted along walkways to form shaded corridors.

COOL PAVEMENT COATINGS ON WALKWAYS

Reflective coatings that reduce the heat absorbed by the pavements.



CLIMATE-CONTROLLED WALKWAYS

Enclosed, air-conditioned walkways that provide stable internal temperature.

CRITERIA CONSIDERED

- THERMAL COMFORT
- COVERAGE & SCALABILITY
- SUSTAINABILITY
- FEASIBILITY

CHOSEN SOLUTION: PARK AND PLANT

AN EXISTING QF COMMUNITY-LED PROGRAM [4]

WHY THIS:

- 12˚C PEDESTRIAN-LEVEL TEMPERATURE REDUCTION[1]
- 60–70% WALKWAY SHADE COVERAGE POSSIBLE[2]
- ZERO ENERGY COST

OUR RECOMMENDATION: INCREASE PLANTING ALONG ALL PRIMARY WALKWAYS.

ADVANTAGES

- CLIMATE-PROOF: COOLS IN ANY HUMIDITY.
- ECO-FRIENDLY: CLEANS THE AIR .
- FEASIBLE: USES EXISTING QF RESOURCES
- SUSTAINABLE: LOW-WATER, NATIVE SPECIES.
- COMPOUNDING: SHADE GROWS YEAR AFTER YEAR.

ALIGNS WITH QNV 2030

- 25% emissions reduction (NDS3)[3]
- Physical activity and well-being[3]
- 60~70% pedestian shade[2]
- Walking over driving[3]

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REFERENCES